OOP Project Report 2567

หัวข้อ: Pizza Website

ชื่อกลุ่ม: Durain

รายชื่อสมาชิก:

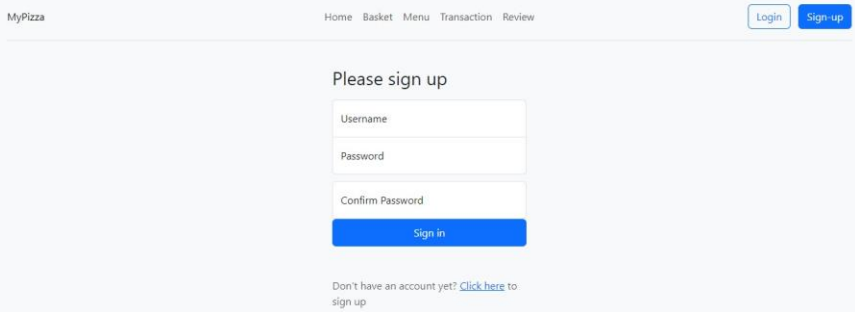
1. พงศภัค ต๊ะต้องใจ 66011428
2. พัฒน์กุลธร ชัยรัตน์ 66011437
3. ราธา โรจน์รุจิพงศ์ 66011464

บทนำ:

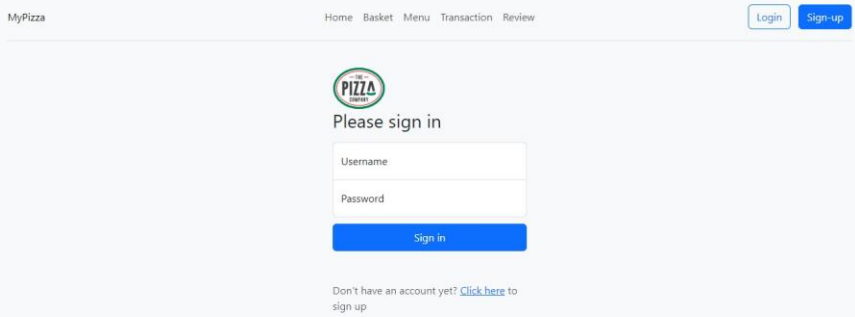
หัวข้อของกลุ่มเราคือการทำเว็บไซต์สั่งพิซซ่าออนไลน์ โดยมีต้นแบบจาก https://1112.com/ เว็บไซต์ของเราจะเริ่มจากหน้า login ก่อน ถ้ากรอก username และ password ถูกต้องก็จะเด้งมาหน้า menu ที่แสดงรายการอาหารชนิดต่างๆอยู่ แต่ถ้าผู้ใช้ยังไม่มีบัญชีก็สามารถ sign up ได้ หลังจากผ่านหน้า login แล้ว ผู้ใช้สามารถมาดูอาหารที่สั่งได้และกดเพิ่มลงตะกร้าตามใจชอบ และเมื่อช็อปจนหนำจำแล้ว ก็สามารถกดยืนยันของในตะกร้าได้ โดยต้องกรอกข้อมูลจัดส่ง เช่น ที่อยู่จัดส่ง หรือรับเองที่ร้าน พร้อมสาขาที่ต้องการ รวมถึงการใช้คูปองลดราคาอาหาร จากนั้นก็จะเด้งไปหน้า confirm payment เพื่อจ่ายเงิน นอกเหนือจากนี้ผู้ใช้แต่ละคนเอง สามารถมาเขียนรีวิวให้กับพิซซ่าได้ด้วย เพื่อรับคูปองส่วนรถเพิ่มเติม ในส่วนของร้านค้าเอง ก็จะมี username และ password สำหรับ role Admin ร้านละหนึ่ง ID โดยทำได้อยู่ 2 อย่าง 1. เช็คสต็อกในแต่ละสาขาว่าเหลือเท่าไหร่บ้าง 2. รีสต็อก เพื่อเพิ่มของในสาขาที่ประจำอยู่ กรณีของใกล้หมด

หน้าจอของ

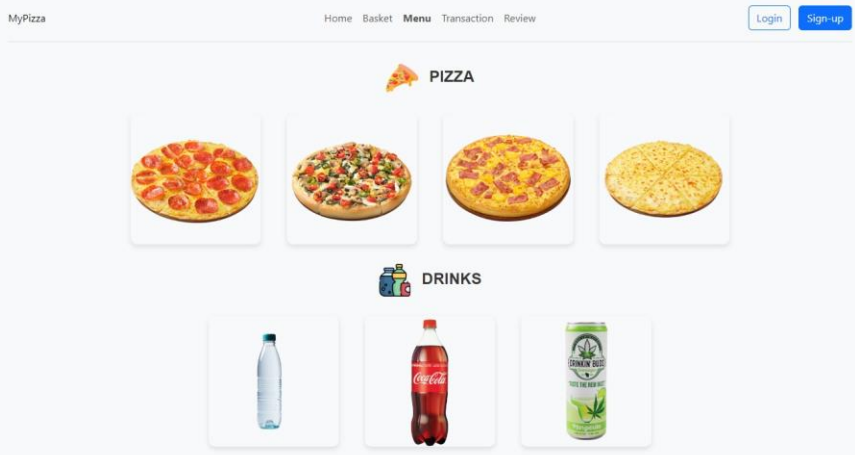
1


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
Please sign up
Username
Password
Confirm Password
Sign in
Don't have an account yet? Click here to sign up

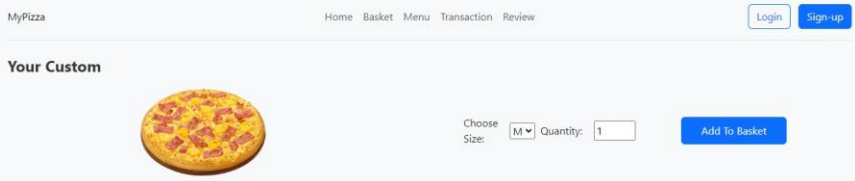
2


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
Please sign in
Username
Password
Sign in
Don't have an account yet? Click here to sign up

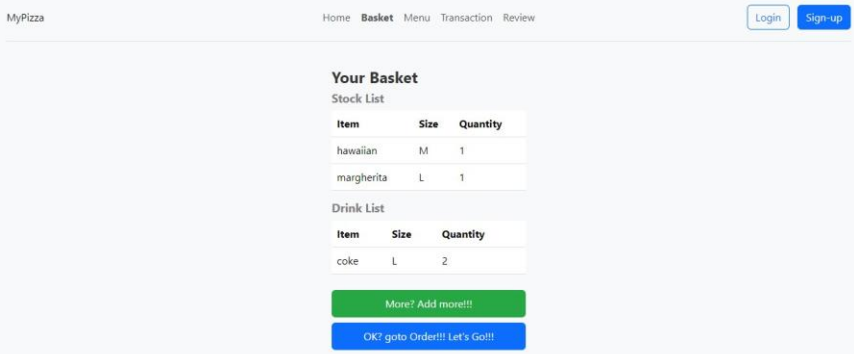
3


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
PIZZA
DRINKS

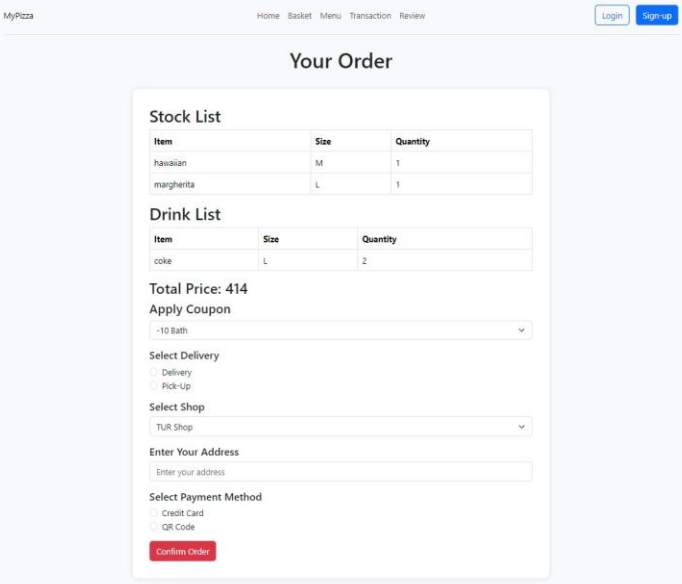
4


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
Your Custom
Choose Size:
M
Quantity:
1
Add To Basket

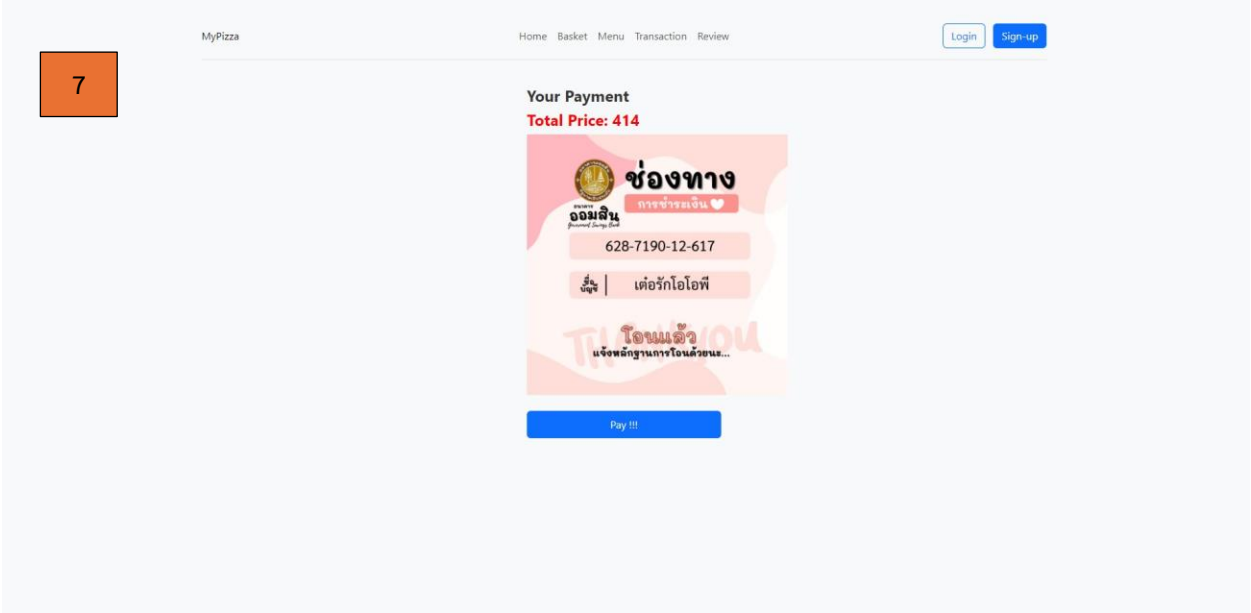
5


MyPizza
Home Basket Menu Transaction Review
Login Sign-up
Your Basket
Stock List
Item Size Quantity
hawaiian M 1
margherita L 1
Drink List
Item Size Quantity
coke L 2
More? Add more!!!
OK? goto Order!!! Let's Go!!!

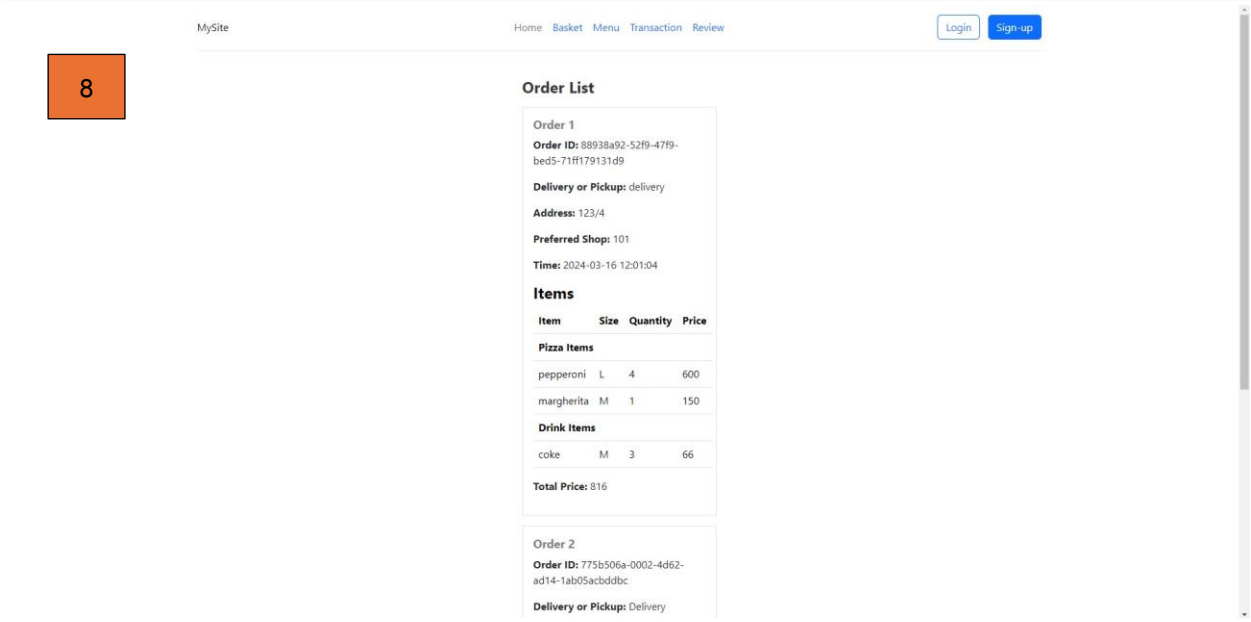
6


MyPizza
Home Basket Menu Transaction Review
Login Sign-up
Your Order
Stock List
Item Size Quantity
hawaiian M 1
margherita L 1
Drink List
Item Size Quantity
coke L 2
Total Price: 414
Apply Coupon
-10 Bath
Select Delivery
Delivery
Pick-Up
Select Shop
TUR Shop
Enter Your Address
Enter your address
Select Payment Method
Credit Card
QR Code
Confirm Order

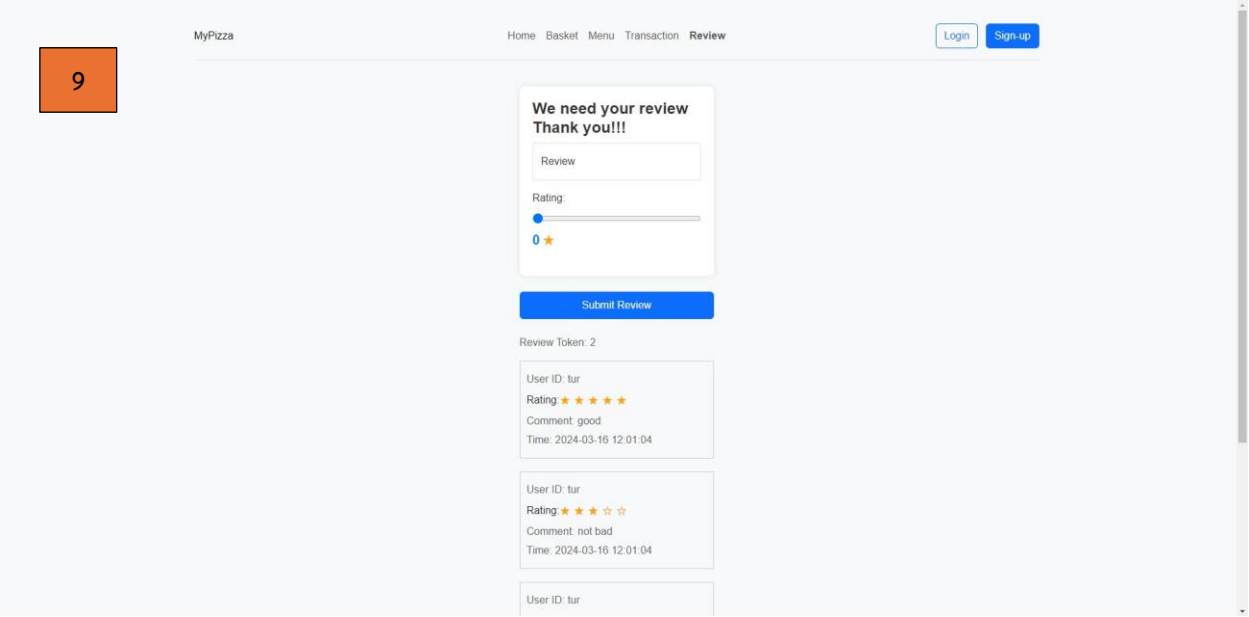
7


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
Your Payment
Total Price: 414
ช่องทาง
628-7190-12-617
เต่อรักโอโอพี
โอนแล้ว
Pay !!!

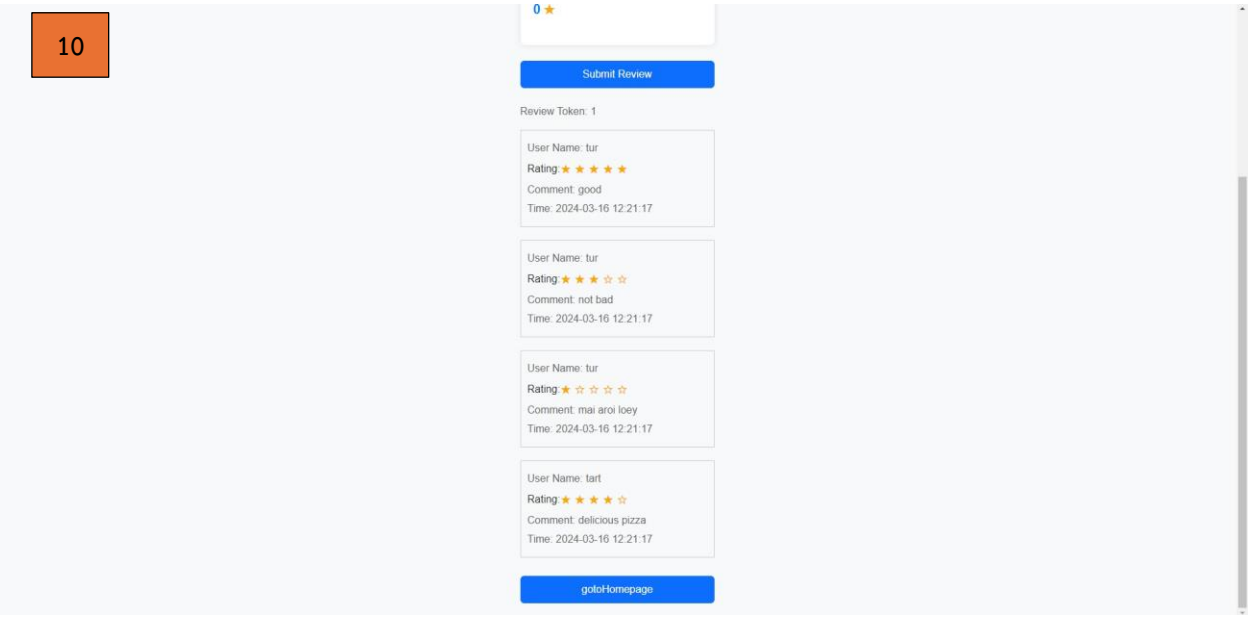
8


MySite
Home Basket Menu Transaction Review
Login
Sign-up
Order List
Order 1
Order ID: 88938a92-52f9-47f9-bed5-71ff179131d9
Delivery or Pickup: delivery
Address: 123/4
Preferred Shop: 101
Time: 2024-03-16 12:01:04
Items
Item Size Quantity Price
Pizza Items
pepperoni L 4 600
margherita M 1 150
Drink Items
coke M 3 66
Total Price: 816
Order 2
Order ID: 775b506a-0002-4d62-ad14-1ab05acbddbc
Delivery or Pickup: Delivery

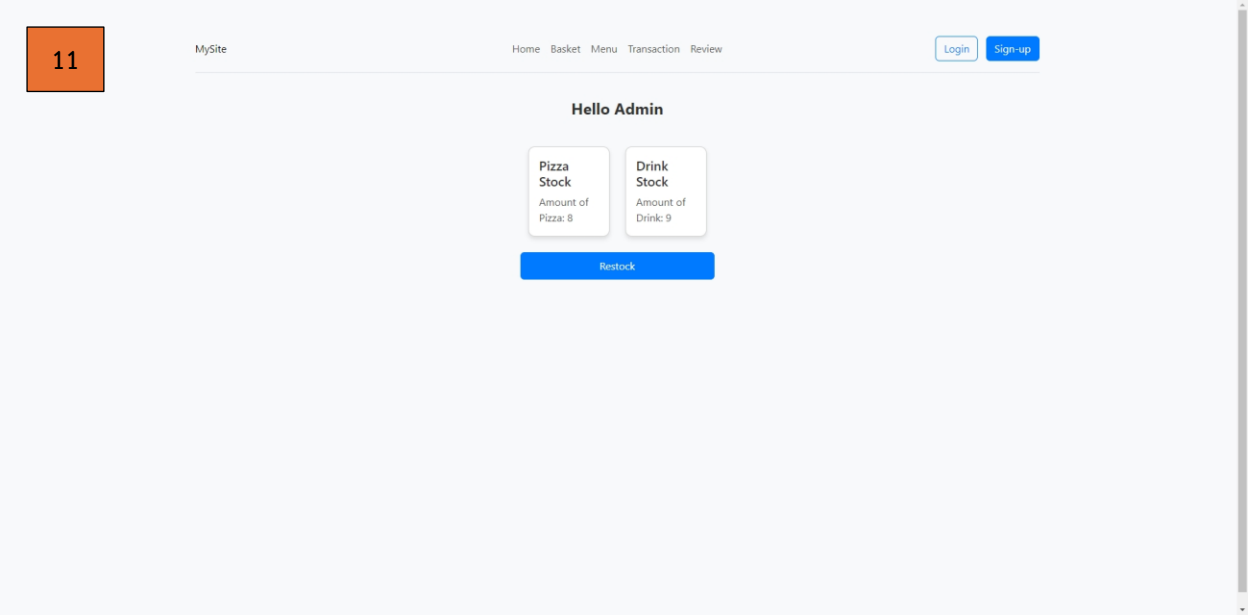
9


MyPizza
Home Basket Menu Transaction Review
Login
Sign-up
We need your review
Thank you!!!
Review
Rating:
0 ★
Submit Review
Review Token: 2
User ID: tur
Rating: ★ ★ ★ ★ ★
Comment: good
Time: 2024-03-16 12:01:04
User ID: tur
Rating: ★ ★ ★ ☆ ☆
Comment: not bad
Time: 2024-03-16 12:01:04
User ID: tur

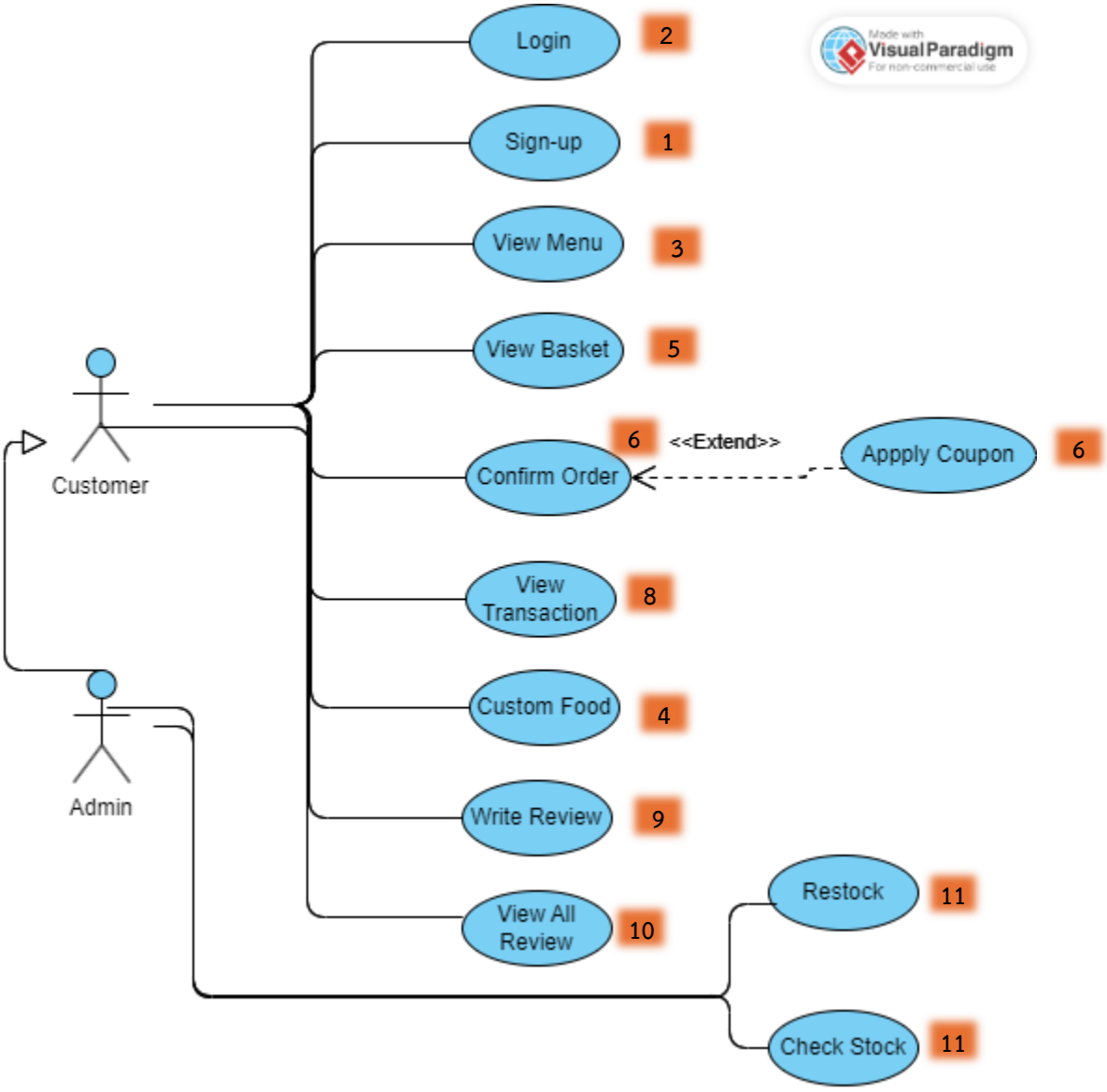
10


0 ★
Submit Review
Review Token: 1
User Name: tur
Rating: ★ ★ ★ ★ ★
Comment: good
Time: 2024-03-16 12:21:17
User Name: tur
Rating: ★ ★ ★ ☆ ☆
Comment: not bad
Time: 2024-03-16 12:21:17
User Name: tur
Rating: ★ ☆ ☆ ☆ ☆
Comment: mai aroi loey
Time: 2024-03-16 12:21:17
User Name: tart
Rating: ★ ★ ★ ★ ☆
Comment: delicious pizza
Time: 2024-03-16 12:21:17
gotoHomepage

11


MySite
Home
Basket
Menu
Transaction
Review
Login
Sign-up
Hello Admin
Pizza Stock
Amount of Pizza: 8
Drink Stock
Amount of Drink: 9
Restock

Use Case Diagram:



Use Case Name	Login
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดปุ่มเข้าสู่ระบบ
Pre-Condition	เข้าหน้าเว็บไซต์
Post-Condition	เข้าสู่ระบบสำเร็จ

Use Case Name	Sign-up
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกรอก username และ password
Pre-Condition	เข้าหน้าเว็บไซต์
Post-Condition	มี user id เป็นของตัวเอง

Use Case Name	View Menu
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดเลือกดูและเลือกอาหารที่ชอบ
Pre-Condition	เข้าสู่ระบบสำเร็จแล้ว
Post-Condition	เลือกรายละเอียดอาหาร (custom)

Use Case Name	Custom Food
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดเลือกรายละเอียดอาหาร (ขนาด, จำนวน)
Pre-Condition	กดเลือกรูปอาหารที่ต้องการ
Post-Condition	อาหารลงตะกร้า และกลับหน้าเมนู

Use Case Name	View Basket
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดดูตะกร้าสินค้าตัวเอง
Pre-Condition	เข้าสู่ระบบสำเร็จแล้ว
Post-Condition	แสดงหน้าตะกร้า

Use Case Name	Confirm Order
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกด confirm order สินค้าในตะกร้า
Pre-Condition	เลือกอาหารที่ต้องการลงตะกร้า
Post-Condition	แสดงหน้า confirm order

Use Case Name	Apply Coupon
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดคูปองที่ต้องการจะใช้ลดราคาอาหาร
Pre-Condition	ต้องมีคูปองอยู่แล้ว ถึงจะกดได้
Post-Condition	ลดราคาอาหารสำเร็จ

Use Case Name	View Transaction
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดดู Transaction ที่เคยสั่งอาหารไปทั้งหมด
Pre-Condition	เข้าสู่ระบบสำเร็จ
Post-Condition	แสดงหน้า Transaction

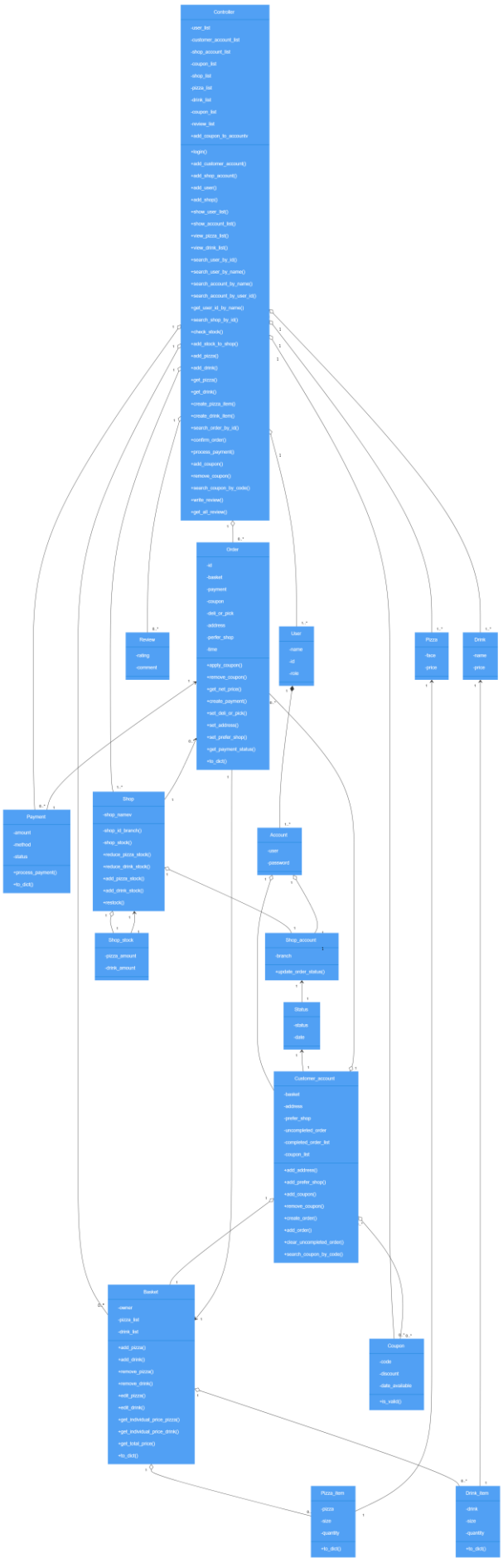
Use Case Name	Write Review
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานเขียนคอมเม้นให้พิซซ่า
Pre-Condition	เข้าสู่ระบบสำเร็จ และเคยสั่งอาหารมาก่อน (1 order ต่อ 1รีวิว)
Post-Condition	รีวิวสำเร็จ

Use Case Name	View All Review
Actors	ผู้ใช้ทั่วไปและผู้ดูแลระบบ
Descriptions	ผู้ใช้งานกดดูรีวิวทั้งหมดที่เคยมีมา ของทุกผู้ใช้งาน
Pre-Condition	เข้าสู่ระบบสำเร็จ
Post-Condition	แสดงหน้ารีวิว

Use Case Name	Restock
Actors	ผู้ดูแลระบบ
Descriptions	ผู้ดูแลระบบกดเติมสินค้าเข้าร้านค้าที่ประจำอยู่
Pre-Condition	เข้าสู่ระบบสำเร็จ
Post-Condition	เติมสินค้าลงร้านค้าสำเร็จ

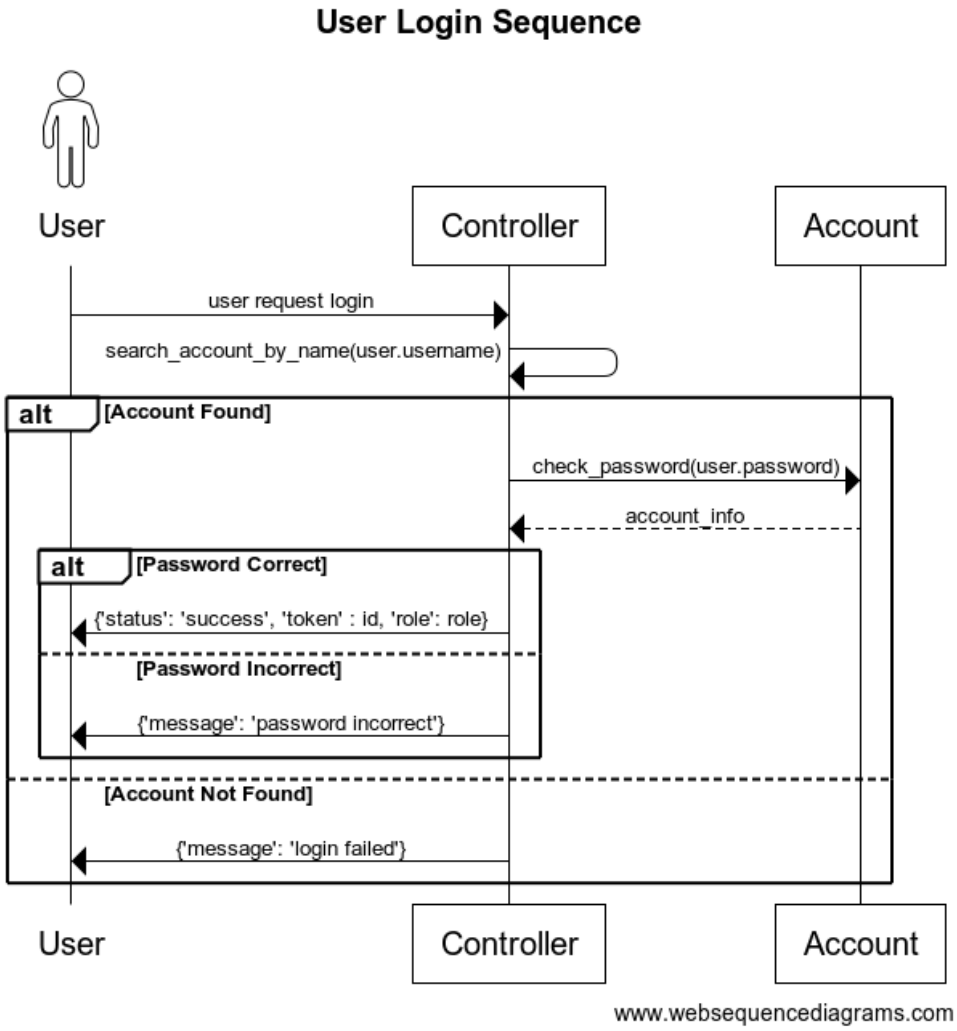
Use Case Name	Check Stock
Actors	ผู้ดูแลระบบ
Descriptions	ผู้ดูแลระบบกดดูจำนวนสินค้าที่คงเหลืออยู่ ณ ร้านค้าที่ประจำอยู่
Pre-Condition	เข้าสู่ระบบสำเร็จ
Post-Condition	แสดงจำนวนสินค้า

Class Diagram:



Sequence Diagram:

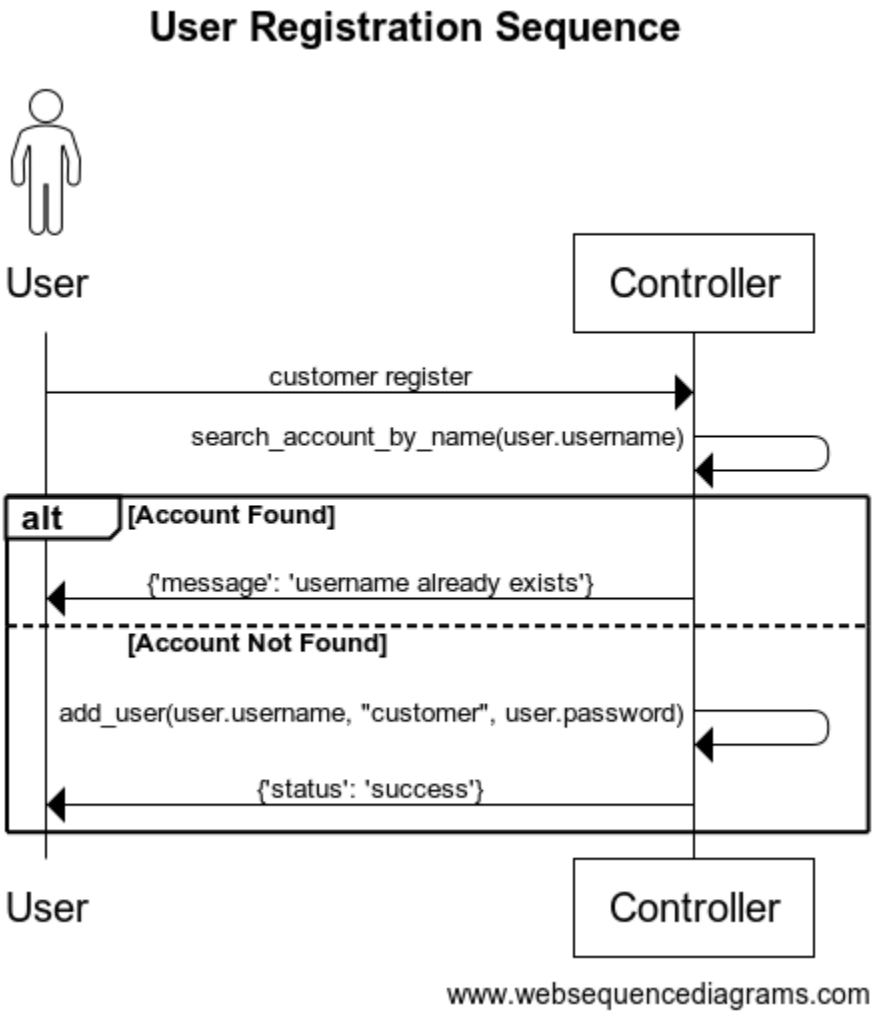
1. เข้าสู่ระบบ (Login)




```python
@app.post('/login/')
async def login(user:loginDTO) -> dict:
    account = controller.search_account_by_name(user.username)
    if account:
        if account.password == user.password:
            id = account.user.id
            role = account.user.role
            return {'status': 'success', 'token' : id, 'role': role}
        else:
            return {"message" : "password incorect"}
    else:
        return {"message" : "login failed" }
```

```python
def search_account_by_name(self, name: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.name == name:
            return account
    return None
```

2. ลงทะเบียน (Register)

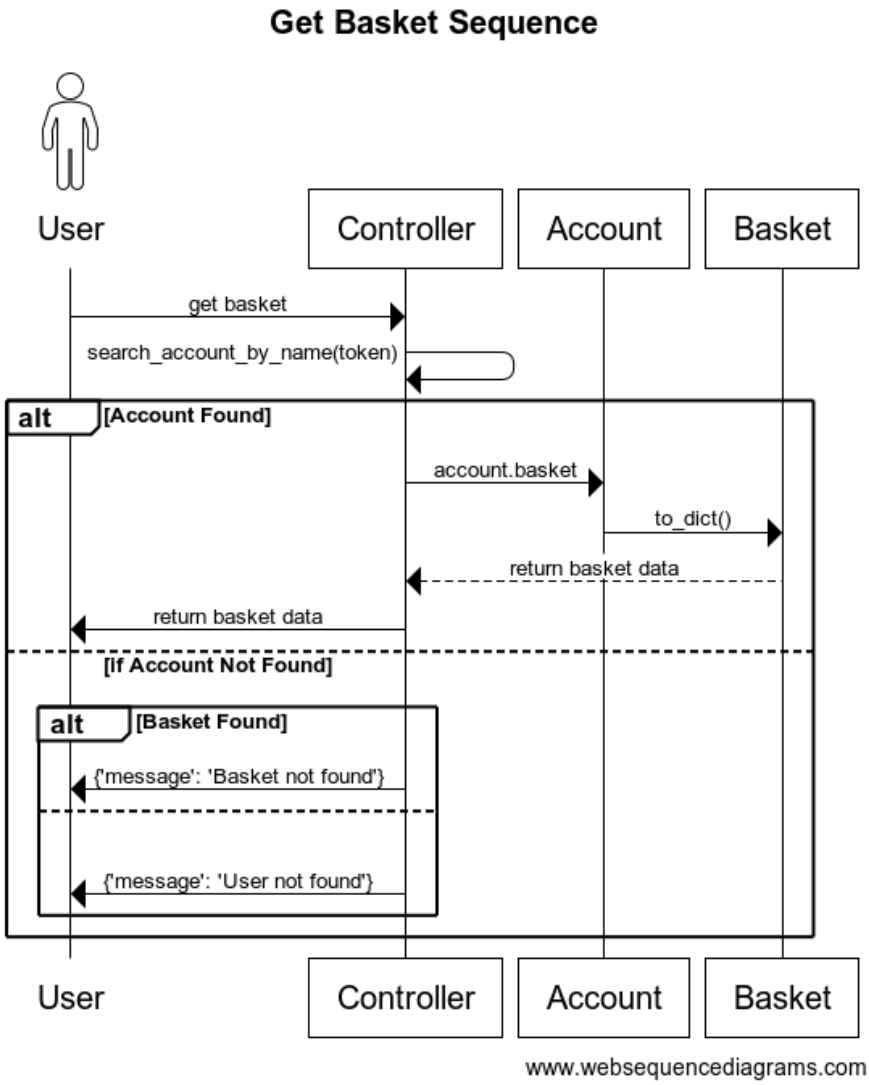



```python
@app.post('/register/', tags=["user"])
async def register(user:loginDTO) -> dict:
    if controller.search_account_by_name(user.username):
        return {"message" : "user name already exist" }
    controller.add_user(user.username, "customer", user.password)
    return {'status': 'success'}
```

```python
def search_account_by_name(self, name: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.name == name:
            return account
    return None
```

```python
def add_user(self, name: str, role: str, password: str) -> None:
    self.__user_list.append(User(name, role))
    if role == "customer":
        self.add_customer_account(name, password)
    elif role == "shop":
        self.add_shop_account(name, password)
```

3. ดูของในตะกร้า (Get basket)

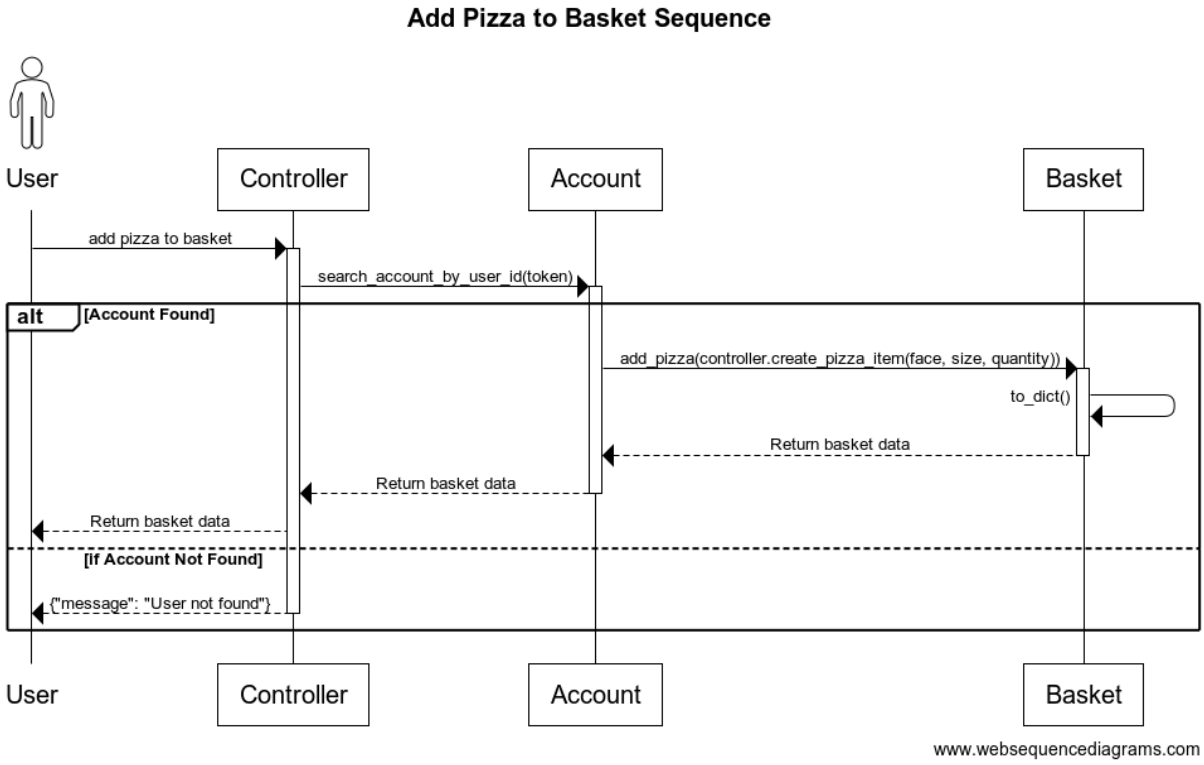



```python
@app.get('/basket/', tags=["basket"])
def get_basket(token: str) -> dict:
    account = controller.search_account_by_name(token)
    # account = controller.search_account_by_user_id(token)
    if account != None :
        if account.basket :
            return account.basket.to_dict()
        else:
            return {"message" : "Basket not found"}
    return {"message" : "User not found"}
```

```python
def search_account_by_name(self, name: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.name == name:
            return account
    return None
```

```python
def to_dict(self) -> dict:
    pizza_list = [{"item": pizza_item.to_dict(), "price": self.get_individual_price_pizza(pizza_item)} for pizza_item in self.__pizza_list]
    drink_list = [{"item": drink_item.to_dict(), "price": self.get_individual_price_drink(drink_item)} for drink_item in self.__drink_list]
    return {"pizza_list": pizza_list, "drink_list": drink_list}
```

4. เพิ่มพิซซ่าลงตะกร้า (Add pizza to basket)



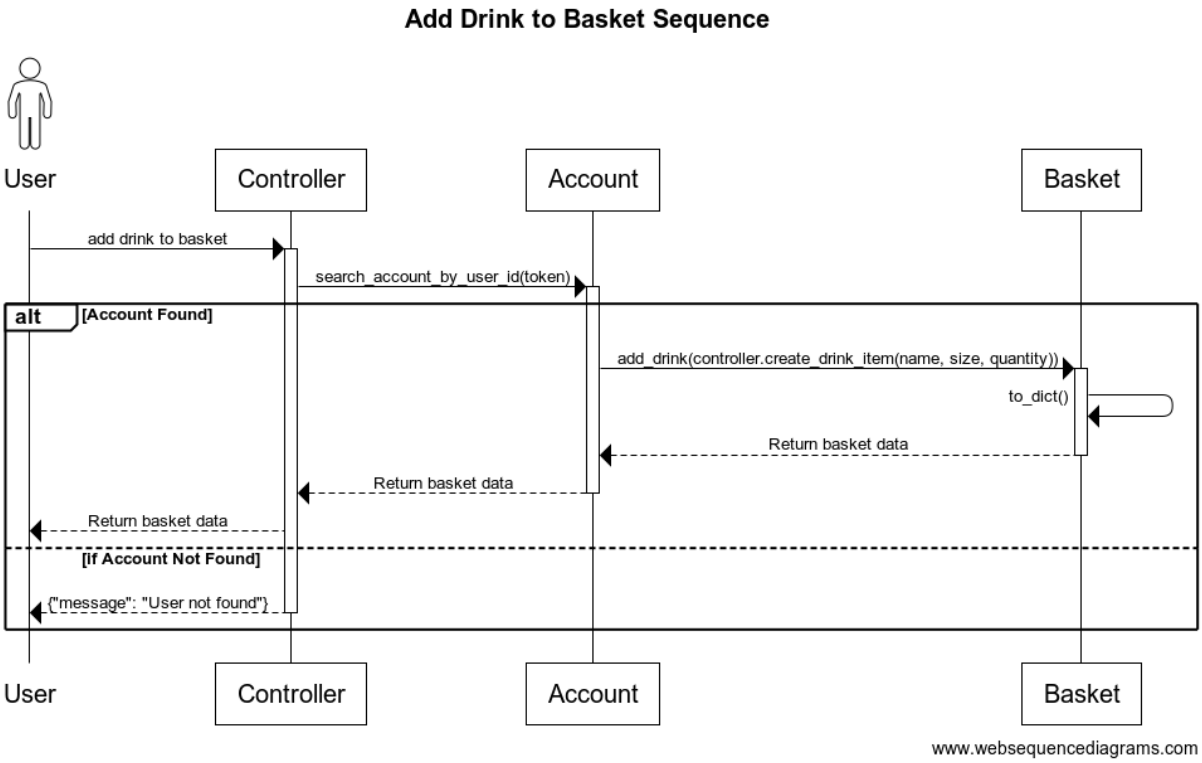

```python
@app.post('/basket/add_pizza/', tags=["basket"])
def add_pizza_to_basket(token: str, face: str, size: str, quantity: int) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    account.basket.add_pizza(controller.create_pizza_item(face, size, quantity))
    return account.basket.to_dict()
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def add_pizza(self, item: Pizza_item) -> None:
    for pizza in self.__pizza_list:
        if item.pizza == pizza.pizza and item.size == pizza.size:
            pizza.quantity += item.quantity
            break
    else:
        self.__pizza_list.append(item)
```

```python
def to_dict(self) -> dict:
    pizza_list = [{"item": pizza_item.to_dict(), "price": self.get_individual_price_pizza(pizza_item)} for pizza_item in self.__pizza_list]
    drink_list = [{"item": drink_item.to_dict(), "price": self.get_individual_price_drink(drink_item)} for drink_item in self.__drink_list]
    return {"pizza_list": pizza_list, "drink_list": drink_list}
```

5. เพิ่มเครื่องดื่มลงตะกร้า (Add drink to basket)



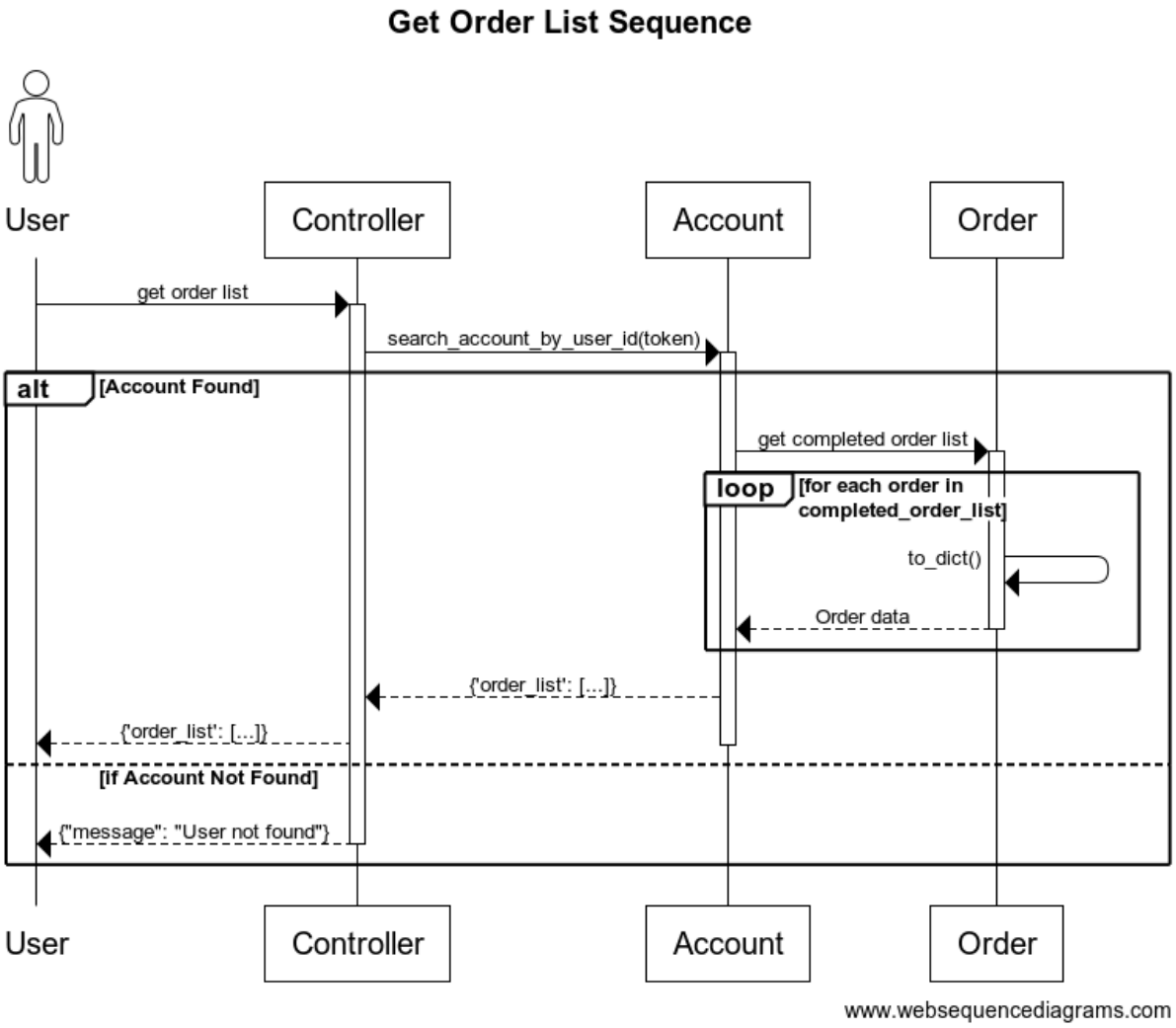

```python
@app.post('/basket/add_drink/', tags=["basket"])
def add_drink_to_basket(token: str, name: str, size: str, quantity: int) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    account.basket.add_drink(controller.create_drink_item(name, size, quantity))
    return account.basket.to_dict()
```

```python
def add_drink(self, item: Drink_item) -> None:
    for drink in self.__drink_list:
        if item.drink == drink.drink and item.size == drink.size:
            drink.quantity += item.quantity
            break
    else:
        self.__drink_list.append(item)
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def to_dict(self) -> dict:
    pizza_list = [{"item": pizza_item.to_dict(), "price": self.get_individual_price_pizza(pizza_item)} for pizza_item in self.__pizza_list]
    drink_list = [{"item": drink_item.to_dict(), "price": self.get_individual_price_drink(drink_item)} for drink_item in self.__drink_list]
    return {"pizza_list": pizza_list, "drink_list": drink_list}
```

6. ดูอาหารที่เคยสั่งทั้งหมด (Get order list)



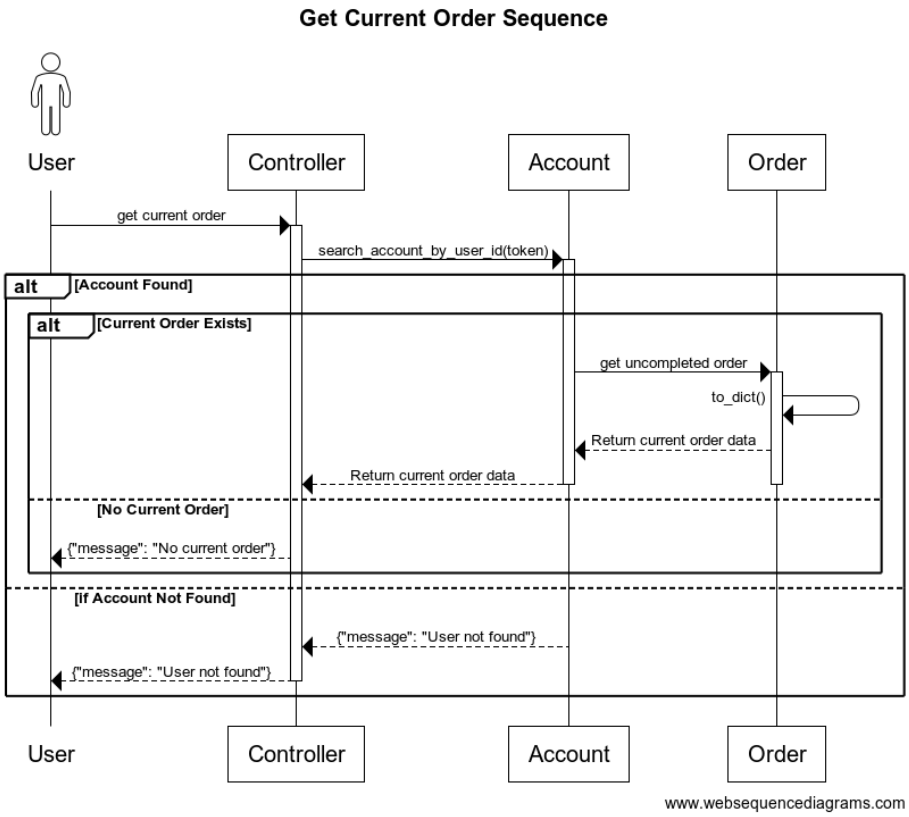

```python
@app.get('/order/', tags=["order"])
def get_order_list(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    return {'order_list':
            [order.to_dict() for order in account.completed_order_list]}
```

```python
def to_dict(self):
    return {
        "order_id": self.__id,
        "basket": self.__basket.to_dict(),
        "coupon": self.__coupon.code if self.__coupon else None,
        "price": self.get_net_price(),
        "deli_or_pick": self.__deli_or_pick,
        "address": self.__address,
        "perfer_shop": self.__perfer_shop,
        "time": self.__time
    }
```

```python
@app.get('/order/', tags=["order"])
def get_order_list(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    return {'order_list':
            [order.to_dict() for order in account.completed_order_list]}
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

7. ดูออเดอร์อาหารของปัจจุบัน (Get current order)

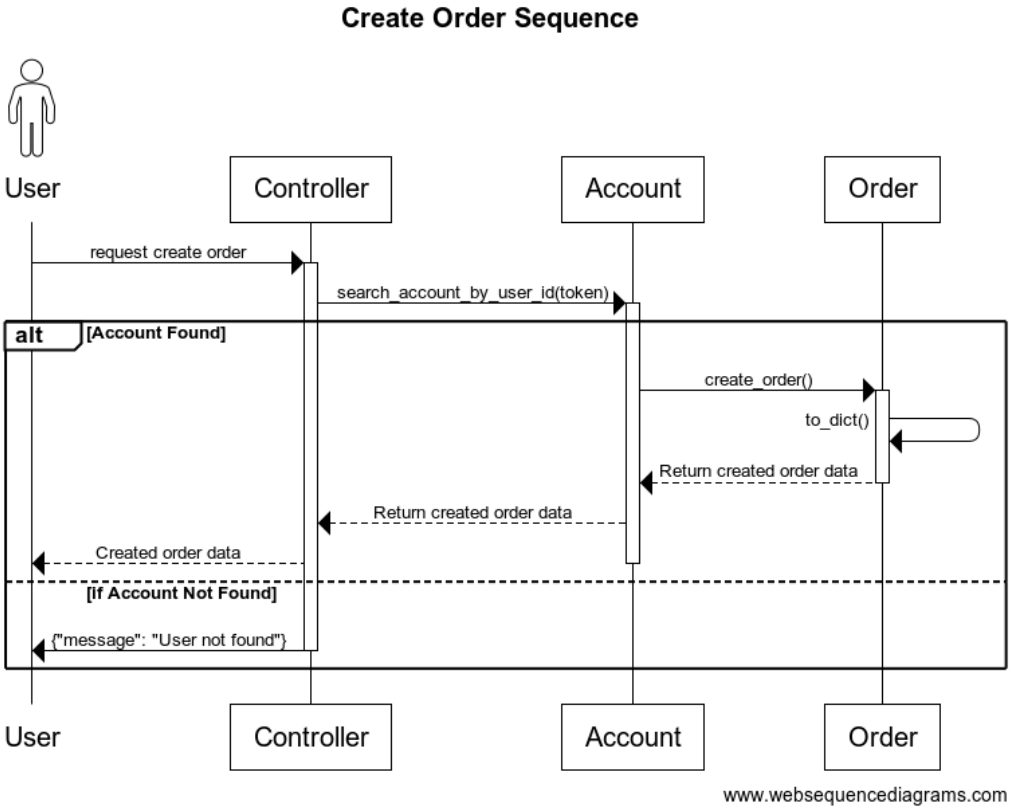



```python
@app.get('/current_order/', tags=["order"])
def get_current_order(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    if account.uncompleted_order == None:
        return {"message" : "No current order"}
    return account.uncompleted_order.to_dict()
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def to_dict(self):
    return {
        "order_id": self.__id,
        "basket": self.__basket.to_dict(),
        "coupon": self.__coupon.code if self.__coupon else None,
        "price": self.get_net_price(),
        "deli_or_pick": self.__deli_or_pick,
        "address": self.__address,
        "perfer_shop": self.__perfer_shop,
        "time": self.__time
    }
```

8. สร้างออเดอร์อาหาร (Crete order)

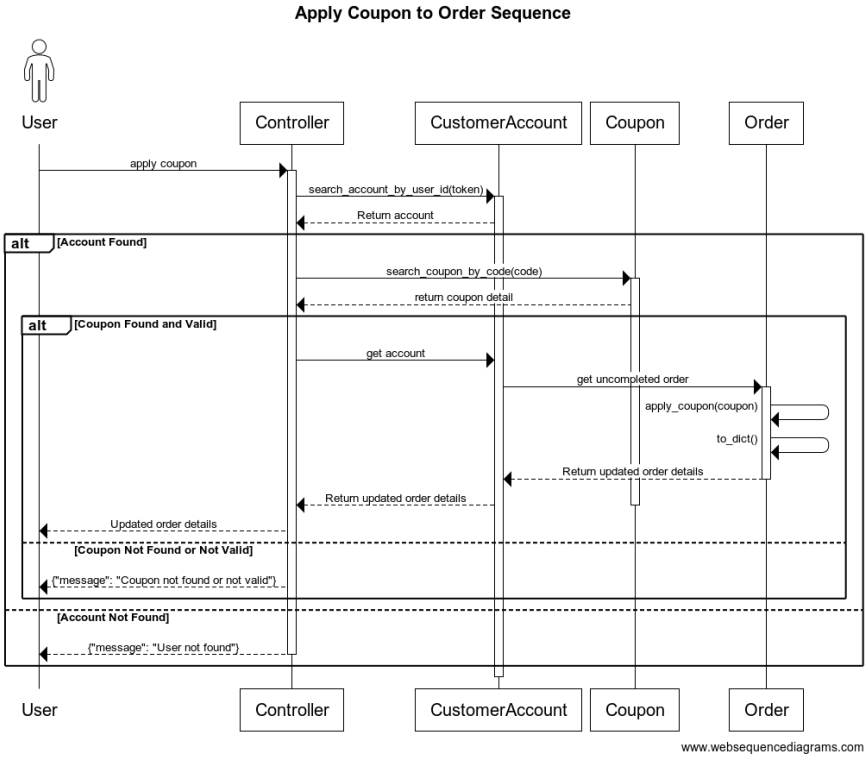



```python
@app.get('/current_order/', tags=["order"])
def get_current_order(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    if account.uncompleted_order == None:
        return {"message" : "No current order"}
    return account.uncompleted_order.to_dict()
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def to_dict(self):
    return {
        "order_id": self.__id,
        "basket": self.__basket.to_dict(),
        "coupon": self.__coupon.code if self.__coupon else None,
        "price": self.get_net_price(),
        "deli_or_pick": self.__deli_or_pick,
        "address": self.__address,
        "perfer_shop": self.__perfer_shop,
        "time": self.__time
    }
```

9. ใช้งานคูปอง (Apply coupon)




```python
@app.post('/order/apply_coupon/', tags=["order"])
def apply_coupon(token: str, code: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    coupon = controller.search_coupon_by_code(code)
    if coupon == None or coupon.is_valid() == False:
        return {"message" : "Coupon not found or not valid"}
    account.uncompleted_order.apply_coupon(coupon)
    return account.uncompleted_order.to_dict()
```

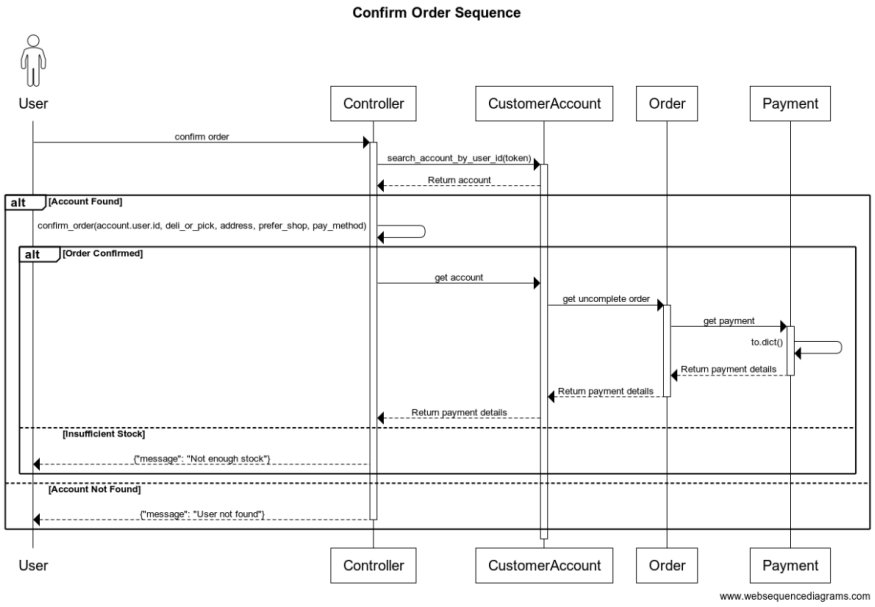
```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def search_coupon_by_code(self, code: str) -> Coupon:
    for coupon in self.__coupon_list:
        if coupon.code == code:
            return coupon
    raise None
```

```python
def is_valid(self) -> bool:
    return self.__date_available > time.strftime("%Y-%m-%d")
```

```python
def to_dict(self):
    return {
        "order_id": self.__id,
        "basket": self.__basket.to_dict(),
        "coupon": self.__coupon.code if self.__coupon else None,
        "price": self.get_net_price(),
        "deli_or_pick": self.__deli_or_pick,
        "address": self.__address,
        "perfer_shop": self.__perfer_shop,
        "time": self.__time
    }
```

10. ยืนยันออเดอร์อาหาร (Confirm order)



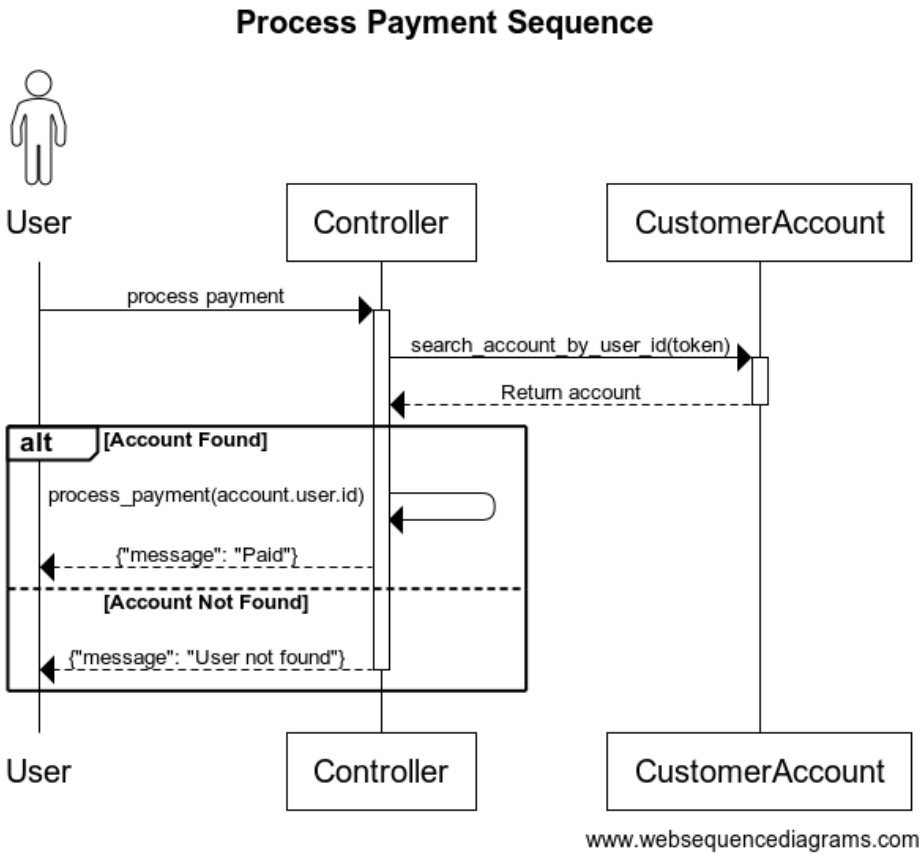

```python
@app.post('/order/confirm_order/', tags=["order"])
def confirm_order(token: str, deli_or_pick: str, address: str, prefer_shop: int, pay_method: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}

    if controller.confirm_order(account.user.id, deli_or_pick, address, prefer_shop, pay_method) :
        return account.uncompleted_order.payment.to_dict()
    else :
        return {"message" : "Not enough stock"}
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def confirm_order(self, user_id: str,deli_or_pick: str, address: str, prefer_shop: int, pay_method: str) -> None:
    account = self.search_account_by_user_id(user_id)
    account.add_address(address)
    account.add_prefer_shop(prefer_shop)
    account.uncompleted_order.set_address(address)
    account.uncompleted_order.set_prefer_shop(prefer_shop)
    account.uncompleted_order.set_deli_or_pick(deli_or_pick)

    pizza_amount = len(account.uncompleted_order.basket.pizza_list)
    drink_amount = len(account.uncompleted_order.basket.drink_list)

    shop = self.search_shop_by_id(prefer_shop)
    pizza_in_stock = shop.shop_stock.pizza_amount
    drink_in_stock = shop.shop_stock.drink_amount

    if pizza_amount > pizza_in_stock or drink_amount > drink_in_stock:
        return False

    shop.shop_stock.set_pizza_amount(pizza_in_stock - pizza_amount)
    shop.shop_stock.set_drink_amount(drink_in_stock - drink_amount)

    account.uncompleted_order.create_payment(pay_method)
    return True
```

```python
def to_dict(self):
    return {
        "amount": self.__amount,
        "method": self.__method,
        "status": self.__status
    }
```

11. คำนวณการจ่ายเงิน (Process payment)

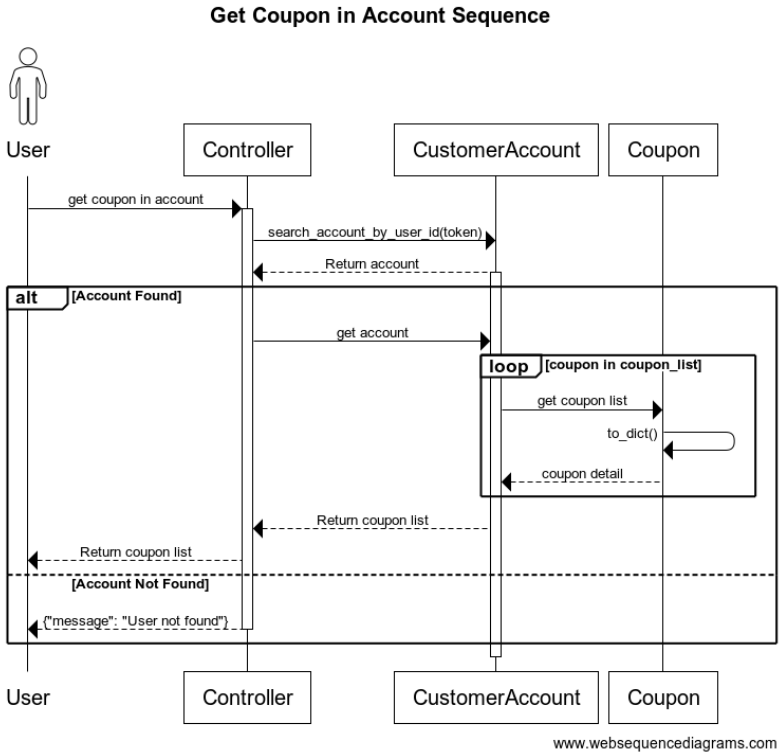



```python
@app.post('/order/process_payment/', tags=["payment"])
def process_payment(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    controller.process_payment(account.user.id)
    return {"message" : "Paid"}
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def process_payment(self, user_id: str):
    account = self.search_account_by_user_id(user_id)
    account.add_order(account.uncompleted_order)
    account.add_review_token(1)
    account.uncompleted_order.payment.process_payment()

    account.clear_basket()
    account.clear_uncompleted_order()
```

12. ดูคูปองของผู้ใช้งานนั้นๆ (Get coupon in account)

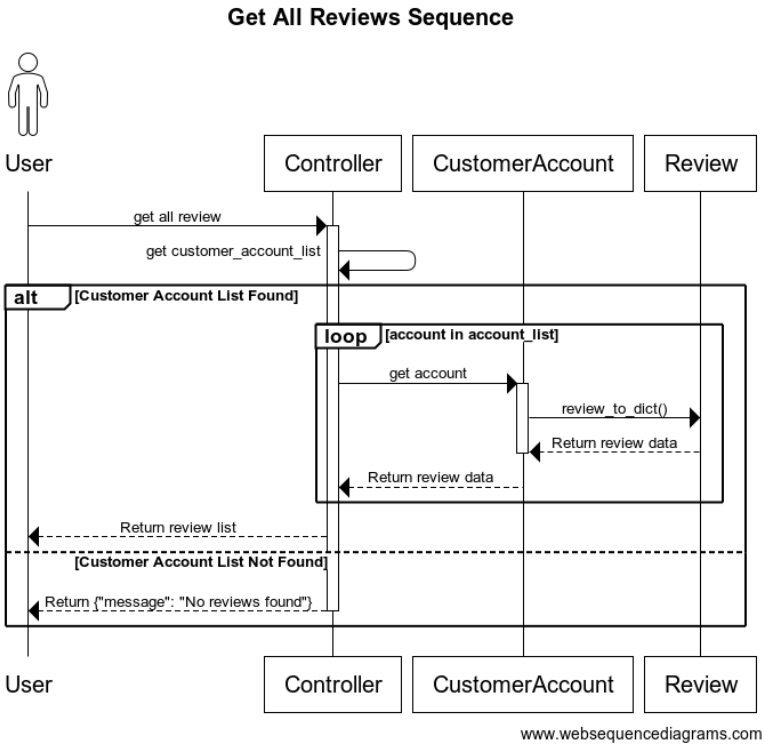



```python
@app.get('/coupon/', tags=["coupon"])
def get_coupon_in_account(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    return {'coupon_list' : [coupon.to_dict() for coupon in account.coupon_list]}
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def to_dict(self) -> dict:
    return {
        "code": self.__code,
        "discount": self.__discount,
    }
```

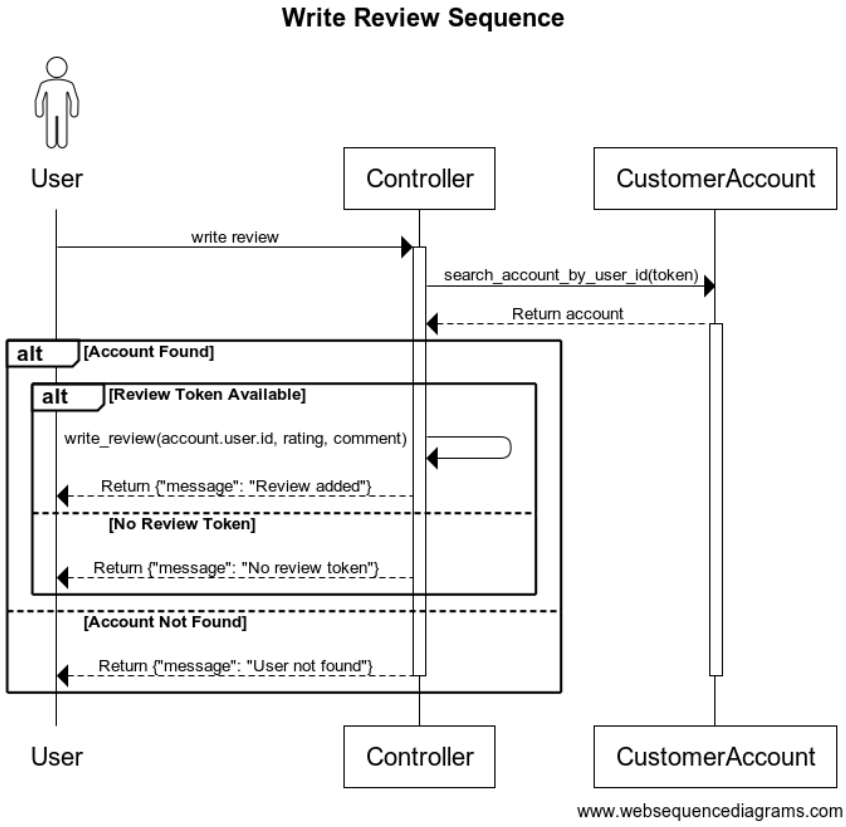
13. ดูรีวิวทั้งหมด (Get all review)




```python
@app.get('/review/', tags=["review"])
def get_all_review() -> dict:
    return {'review_list': [account.review_to_dict() for account in controller.customer_account_list]}
```

```python
def review_to_dict(self) -> list:
    return [{'user_id': self.user.name,
             'rating': review.rating,
             'comment': review.comment,
             'time': review.time}
            for review in self.__review_list]
```

14. เขียนรีวิว (Write review)

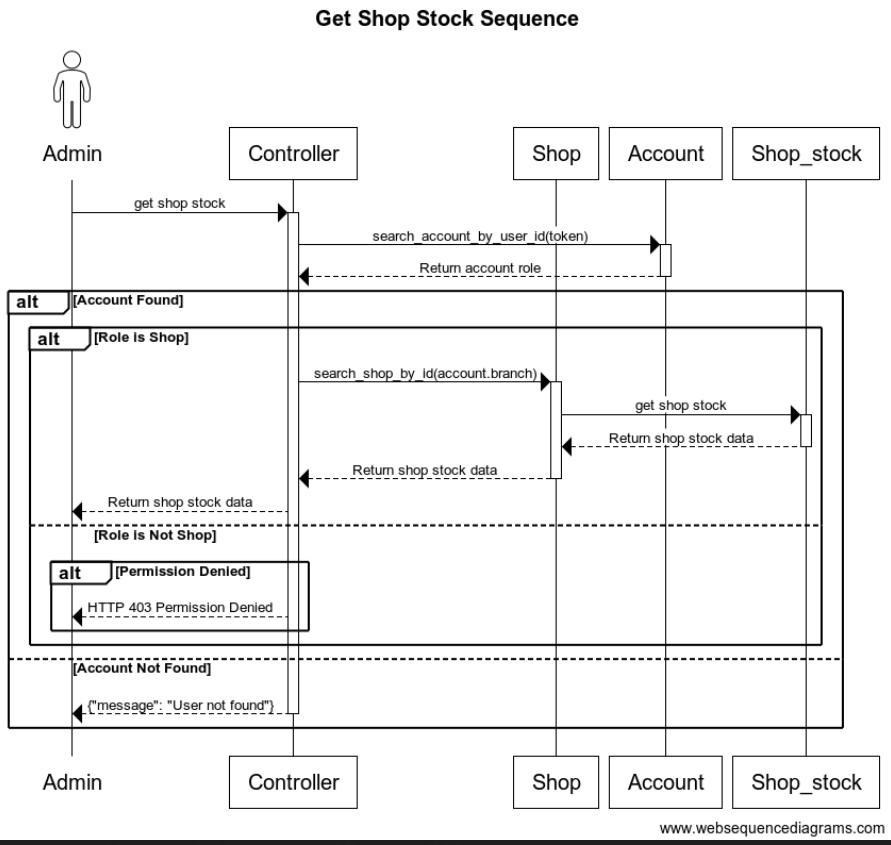



```python
@app.post('/review/', tags=["review"])
def write_review(token: str, rating: int, comment: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    if account.review_token == 0:
        return {"message" : "No review token"}
    controller.write_review(account.user.id, rating, comment)
    return {"message" : "Review added"}
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def write_review(self, user_id: str, rating: int, comment: str ) -> None:
    account = self.search_account_by_user_id(user_id)
    if account and account.review_token > 0:
        account.add_review_token(-1)
        review = Review(rating, comment)
        account.add_review(review)
    else:
        return None
```

15. ดูคลังสินค้า (Get shops stock [Only Admin])

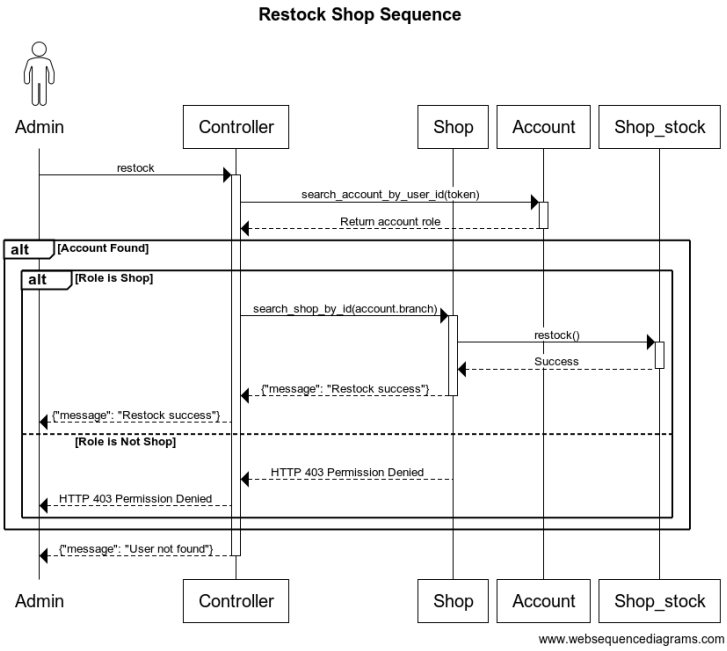



```python
@app.get('/stock/', tags=["stock"])
def get_shops(token: str) -> dict :
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    if account.user.role != "shop":
        raise HTTPException(status_code=403, detail="Permission denied")
    shop = controller.search_shop_by_id(account.branch)
    return shop.shop_stock.to_dict()
```

```python
def search_account_by_user_id(self, user_id: str) -> Account:
    for account in self.__customer_account_list + self.__shop_account_list:
        if account.user.id == user_id:
            return account
    return None
```

```python
def search_shop_by_id(self, shop_id_branch: int) -> Shop:
    for shop in self.__shop_list:
        if shop.shop_id_branch == shop_id_branch:
            return shop
    return None
```

16. เติมคลังสินค้า (Restock [Only Admin])




```python
@app.post('/stock/restock/', tags=["stock"])
def restock(token: str) -> dict:
    # account = controller.search_account_by_name(token)
    account = controller.search_account_by_user_id(token)
    if account == None :
        return {"message" : "User not found"}
    if account.user.role != "shop":
        raise HTTPException(status_code=403, detail="Permission denied")
    shop = controller.search_shop_by_id(account.branch)
    shop.restock()
    return {"message" : "Restock success"}
```

```python
def search_shop_by_id(self, shop_id_branch: int) -> Shop:
    for shop in self.__shop_list:
        if shop.shop_id_branch == shop_id_branch:
            return shop
    return None
```

```python
def restock(self) -> None:
    self.__shop_stock.set_pizza_amount(Shop_stock.dialy_stock)
    self.__shop_stock.set_drink_amount(Shop_stock.dialy_stock)
```